SkillPilot Architecture Handbook · Volume Ia

Skill Graph Architecture, Media, Goal Book, and Package Interchange

Layered Goal System, Node Types, View Projection, Source and Content Integration,

Visualizations, Learning-Goal Book Pipeline, Gymnasium Rollout, and Dual Package Releases

Scope of this volume

Complementing the formal Graph Definition (Volume I), Volume Ia consolidates the architectural, media, publication, and interchange contracts surrounding the skill graph: the layered general goal system and incremental migration, pedagogical node types, deterministic view projection and goal-placement semantics, source vs. external content-package separation (Issue #50), human-readable source rationales, atomic goal visualizations, the learning-goal book and multi-AI evidence-review pipeline, the canonical German Gymnasium rollout policy, and MEM/FWU ontology roundtrips plus dual curriculum package releases.

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1 General Goal System and Incremental Migration

SkillPilot's long-term goal model follows two governing principles: **"As much semantics as necessary, as little ontology as possible"** and **"Goals are stable. Structure is contextual."** The formal mathematical graph core is specified in Graph Definition; projection rules live in 3, and runtime fallback mechanics live in Minimal Schema & Runtime Fallback.

1.1 Four Semantic Layers Plus One Composition-View Layer

Rather than mixing subject competencies, school-year containers, and process taxonomies into one undifferentiated hierarchy, the target architecture separates four semantic layers plus one learner-facing projection layer:

Layer	JSON / Artifact Carrier	Role and Invariants
Goal layer	`goals[]`, `contains`, `requires`	Durable assessable atomic goals and subject-matter clusters; owns mastery, frontier eligibility, and canonical IDs.
Program layer	`programUnits[]` (`id`, `kind`, `label`, `order`, `parentUnitId`)	Contextual structural units such as year, semester, module, phase, track, or exam, and learning units (`kind: "learningUnit"`, fields in Graph Definition §11.2); never mastered directly.
Placement layer	`goalPlacements[]` (`goalId`, `unitId`, `relation`, `context`)	Attaches unchanged canonical goals (`primary`, `secondary`, `assessed`) to program units under explicit scope contexts; `own` assigns an atomic goal, without context, to the learning unit that owns it directly.
Competency-axis layer	`competencyCatalog[]` (`id`, `label`, `dimension`), `competencyRefs[]`	Orthogonal capability taxonomies (for example `K1`–`K6` process competencies); never main content parents unless assessable.
Composition-view layer	`*.view.json` (`structure`, `canonicalSubtree`, `goal Entry`)	Scope-specific learner-facing upper trees referencing reviewed canonical subtrees without duplicating goal payloads.

Interpretation invariants:

- Year, semester, module, and phase containers are **not** the durable semantic backbone of the skill graph; `phase` is retained as compatibility/badge metadata rather than canonical parentage.
- Broad capability families (`K1`–`K6`) are taxonomy axes, not program units; only concrete, assessable process skills become actual `LearningGoal` nodes.
- Initial learner narrowing (jurisdiction, school form, stage, duration model, course profile) acts as **entry scope** over program units, placements, composition views, and compiled `applicability` (Levels of Personalization), never as duplicated canonical goals.

1.2 Recursive Learning Units and Incremental Migration

Recursive educational units (Graph Definition §11, Learning Units; §12, Optional Planning) compose over the same stable canonical goals:

1. Within one resolved course, each planned atomic acquisition has **one directly owning unit**; ancestor units aggregate child targets without re-owning them.
2. Competence mastered or recognized in a child unit is **globally effective** immediately, even while a broader parent integration exam remains open; local recognition applies only to directly owned ordinary targets inside units that satisfy the validity rules R1–R5 of 11 November 2026 (Graph Definition §13.3).
3. Passing a parent unit's integration exam never marks protected child exams or memorization decks as passed; explicit whole-unit progression (Unit Progression) is validated separately from atomic `requires` acyclicity.

> [!NOTE] **Implementation boundary:** This chapter defines the target semantic split and incremental 6-phase migration path (additive optional schema fields → projection-first UI with `phase` fallback → pilot slice → subtree adoption → composition-view cutover → legacy retirement) without forcing a big-bang rewrite or claiming an already implemented `LearningUnit` runtime compiler.

2 Node Types

SkillPilot classifies learning goals into four pedagogical goal kinds that are orthogonal to the structural atomic/cluster distinction (Graph Definition) and to program-unit kinds (3).

2.1 Outcome Contract and Compiled Applicability

Node Type	Pedagogical Purpose	Typical Content & Completion Basis
Orientation	Meaningful, low-pressure entry into a subject or phase	Concrete possibilities, personal interest follow-up, and explicit learner closure (`semanticKind: "orientation"`, rule `GVR-014`).
Understanding	Conceptual or procedural subject mastery (default)	Assessable `"The learner can ..."` competencies governed by `requires`.
Memorize	Durable recall of facts, vocabulary, and formulas	SM-2 spaced-repetition practice plus a mandatory coach-led **verified hard recall** check over all required cards.
Exam	Realistic scored assessment simulation	Verbatim `examData` tasks graded strictly by `scoring.steps` under neutral AI Exam Mode supervision.

Every atomic goal shares one binary outcome contract: **mastered or not mastered** (persisted as `1` / `1.0` for runtime compatibility; see Learning Workflow). Cluster progress is derived strictly from unique atomic descendants. Optional compiled `applicability` metadata (for example `{"jurisdiction": ["DE-HE", "DE-BY"]}`, with authoring exceptions in `extended Data.applicabilityOverrides`) filters visibility in scoped views without defining a node type or replacing semantic `tags`.

2.2 Orientation, Understanding, and Memorize Semantics

Orientation (`semanticKind: "orientation"`): Checked by `GVR-014` (Graph Validation Rules) to carry zero `requires` and no exam, memory, example, or assessment metadata. Orientation never tests prior knowledge or terminology. Completion follows a four-step dialogue: (1) present possibilities and invite an interest or question, (2) respond to a bare label with a tailored follow-up inviting a personal reaction, (3) summarize positively and offer closure, and (4) persist `1` **only after** the learner explicitly agrees to close (never exposing a graded `0/50/100` UI control). At genuine zero-progress entry into a scope, unresolved orientation goals form the sole selectable frontier; learners with existing progress are never reset to the orientation gate (see Graph Definition §8.2 and didactic routes in §3.3).

Understanding: The default node kind when no specialized payload is present; availability follows `requires` and optional `applicability`.

Memorize (SM-2 + Verified Recall): Each card maintains two process-data lanes that determine the single goal outcome:

- **SRS practice lane:** `interval` (days), `repetition`, `ef` (easiness factor, clamped to ≥ 1.3), and `nextReview` (ms). On review, `quality >= 3` increments `repetition`, sets `interval` to `1`, `6`, or `round(lastInterval * ef)`, updates `ef`, and schedules `now + interval * 24h`; `quality < 3` resets `repetition = 0` and `interval = 1` with unchanged `ef`. Due cards (`next Review <= now` or unseen) are shuffled per session (up to 20 cards). Interactive MCP practice

components translate **known / not known yet** into SM-2 updates; flipping cards never writes state, and completing an SRS practice queue **never** proves mastery.
- **Verified recall lane (`verifiedRecall`):** Tracks `status` (`passed` | `failed`), `attempts`, `failures`, `lastTestedAt`, `passedAt`, and `lastFailedAt`. Learners may trigger **Verify** at any time: the coach fetches only the prompt first, waits for an unassisted answer, then fetches the expected answer and records `passed` (updating SRS as a strong retrieval) or `failed` (re-queuing as due). Controlled recall of **all required cards** in a fully resolved, valid deck establishes permanent goal completion; subsequent SRS due dates never withdraw established goal mastery (the 29 October 2026 code inspection of `LearnerService.java` `isSrsMasteredByVerifiedRecall` and `useSrsMastery.ts` records current flag derivation without authorizing mastery withdrawal).

2.3 Exam Nodes, Recognition Review, and Scoring

Exam nodes embed `examData` (`taskContent`, optional `taskContentEn`, `solutionContent`, optional `solutionContentEn`, and `scoring: {maxPoints, passingPoints, steps: [{id, points, description}]}`). Learners may attempt exams directly without completing preparatory prerequisites first; failed attempts use `requires` for targeted remediation. Under the 4 November 2026 Assessment-Gated Completion concept (implementation adaptation pending), assessments that substantiate global competence recognition must undergo static authoring review proving that no candidate can pass via unrelated compensating points while omitting an essential recognized competence. Under the unit validity rules of 11 November 2026, an assessment claims coverage (`coveredGoalIds`) only for goals it requires directly or transitively (R3), every own ordinary goal of a unit lies on a prerequisite path from an own assessment of that unit (R1), and at least one such assessment carries reviewed evidence that its tasks and normal grading genuinely test the goal (R2); a missing path is a finding for subject review, never a reason to add a redundant edge (Graph Definition §13.3).

Scoring, persistence, and **Exam Mode** supervision follow strict rules:

- **Formulas:** total $= \min(\sum \text{stepPointsAwarded}, \text{scoring.maxPoints})$ and passed $= (\text{total} \geq \text{scoring.passingPoints})$ (logic errors yield `0` step points; arithmetic slips allow partial step points).
- **Binary persistence:** On `passed == true`, set mastery to `1.0` immediately after grading (no extra learner confirmation required; never claim persistence if the write fails). On `passed == false`, leave mastery unchanged and persist **no** failed attempt or score (unlimited retries remain available; revealing the solution afterward is a documented evidential trade-off).
- **AI Exam Supervisor workflow:** (1) Show a neutral 2–4 line exam header, deliver `examData.taskContent` **verbatim** without chunking or hints, and append one fixed submission line; (2) accept only readability clarifications until a single complete submission or explicit give-up arrives; (3) grade strictly against `scoring.steps`; (4) persist mastery on pass only; (5) report score, pass/fail, and step feedback, reveal `solutionContent`, switch back to learning-coach mode, and wait for an explicit learner choice before starting new content.

3 View Projection and Goal Placement

A careful reader must be able to reconstruct every visible learner-facing tree directly from documented JSON fields without inspecting runtime code or relying on title/`phase` heuristics.

3.1 Four View Families and Five Non-Negotiable Projection Rules

View Family	Primary Question Answered	Exclusive Structural Source of Truth
Content tree	What belongs under what in subject-matter terms?	`goals` + explicit `contains` edges (ordered by `contains` array).
Program tree	Where in the curriculum is a goal introduced, revisited, or assessed?	`programUnits` (`parentUnitId`, `order`) + `goalPlacements` (`primary` attaches; `secondary`/`assessed` render as badges/ overlays, missing yields `unplacedInScope`; `own` records learning-unit ownership and attaches nothing).
Competency view	Along which capability or process axis is a goal classified?	`competencyCatalog` + `competencyRefs` (orthogonal taxonomy projection).
Composition tree	How does one resolved scope see the subject as one default tree?	Reviewed `*.view.json` (`structure` + `canonicalSubtree` / `goalEntry`) + canonical `contains` expansion.

Five binding rules govern all projections:

1. **Rule 1 (No heuristic inference):** Never infer parent-child edges from titles, localized labels, `phase`, `dimensionTags.phase`, or string similarity.
2. **Rule 2 (Explicit edge provenance):** Every visible edge originates from exactly one documented field (`contains`, `parentUnitId`, `goalPlacements`, `competencyRefs`, or composition-view `children`).
3. **Rule 3 (Single relation family per view):** Enrich views with badges or filters, never by silently mixing parent hierarchies.
4. **Rule 4 (Reviewed composition view wins):** When a scope-specific `.view.json` exists, it defines the default learner-facing tree while `programUnits`/`goalPlacements` remain valid for program views and diagnostics (`ALL` is a query-only sentinel, never serialized in placements or applicability).
5. **Rule 5 (Single-occurrence tree per resolved scope):** Within one resolved scope (such as `DE-HE / Gymnasium / SekII / LK`), each visible goal appears **at most once** under **at most one** visible parent; pairwise overlapping `canonicalSubtree` expansions are compile-time errors.

3.2 Prerequisite Closure, Personal Learning Scope, and Canonical Identity

Under the Product Owner decisions of **9 November 2026**, **10 November 2026**, and **11 November 2026** (Graph Definition §§5.1–5.2):

- **Unpruned prerequisite closure (9 Nov 2026) & learner-read resilience (11 Nov 2026):** Selecting atomic goals must preserve their complete transitive prerequisite demand on the unpruned canonical graph (`A requires B requires C` retains `B` even if `A` and `C` alone matched initial scope filters). Unknown IDs, cycles, or unaccounted required atoms remain hard errors in validation, curriculum QA, and native acceptance; on learner reads, when a

learning context's canonical graph is not strictly compilable, the runtime keeps the reviewed seed scope without the derived closure, claims no learning basis, offers no goal whose demand is broken, and logs the failure once per curriculum and error code (errors in the seed selection itself remain hard errors).

- **Personal learning-scope basis L (10 Nov 2026):** Necessary prerequisites belong to learning, progress, and completion, not merely hidden lookup support. Given valid reviewed atomic seed targets S, compile the fixed global target basis $L = S \cup \{p \in A \mid \exists t \in S : (t, p) \in R_{\text{at}}^{+}\}$ on the unpruned canonical graph. L depends solely on personal configuration and curriculum/projection revision, **never** on current mastery. Mastered atoms remain in L as already satisfied; open prerequisites receive navigable acquisition targets in the full Personal Curriculum with explainable `required by` provenance (without inventing new source mappings, enrolling unrelated siblings, or mutating mastery). Level 3 focus equals the selected roots' descendant union intersected with L (normal mode masks checks to focus *after* effective transitive reachability on canonical `requires`, without erasing paths through outside-focus intermediates).
- **Subject-local `projectionRole` specificity:** Authored view references carry `projectionRole: "target" | "prerequisiteOnly"` (default `"target"`). Specificity resolves overlaps via (1) direct `goalEntry` over `canonicalSubtree`, (2) deeper/nearest `canonicalSubtree` over broader ancestor subtrees, and (3) `"target"` on equal specificity. A goal marked `prerequisiteOnly` in one subject (for example Hessen Sek-II-LK Mathematics bridge/formula goals or Physics math support) still requires an acquisition target somewhere in the closed Personal Curriculum L if genuinely required.
- **Preserved canonical atomicity:** Collapsing or referencing a canonical cluster never creates an opaque substitute atom and never transfers cluster results to children (see also the historical `DPK-006c-c` boundary in Dual Package Implementation Status and tooling split in Curriculum Authoring Editors).

3.3 Learning Units in Projected Views

Learning units are declared in the placement layer of their skill landscape (Product Owner decision of 12 November 2026): a `programUnits` entry of kind `learningUnit` owns atomic goals through `goalPlacements` of relation `own` and names the unit that requires it as a child through `parentUnitId` (Graph Definition §11.2; field reference); `validate:learning-units` checks these declarations. The program-tree projection attaches goals only through `primary` placements, so an `own` placement changes no visible tree.

Projected Learning Units (target concept; runtime compiler pending) bind to the reviewed composition view or program projection:

1. **Single direct course owner:** Every planned atomic acquisition has exactly one directly owning unit in the resolved course; ancestors aggregate child obligations without duplicating ownership.
2. **Cross-subject identity:** Shared canonical goals (for example `target` in Mathematics and `prerequisiteOnly` in Physics) count once in the joint plan while retaining subject-local roles.
3. **Full prerequisite accounting:** Every required atom must be covered by the explicit entry basis or a planned acquisition no later than its dependent target.
4. **Local recognition & block ordering:** Implicit recognition evaluates only against each directly owning unit's reviewed contract; parent exams neither reopen completed child units nor pass protected child exams/cards, and whole-unit progression remains a separate explicit course policy.

4 Source and Resource Links

SkillPilot enforces a strict inversion principle for external learning media: **"Content references the curriculum; the curriculum does not select providers."**

4.1 Three-Way Purpose Split and Hard Canonical Prohibition

Mechanism	Carrier	Permitted Purpose & Constraints
Official curriculum provenance	`sourceRef` and curriculum-typed `resourceLinks` (`type: "curriculum"`)	Trace canonical goals to official state/ institutional syllabi, statutes, and page locators (6).
First-party goal visualizations	`resourceLinks` with `type: "goal-visualization"`	Optional curated 16:9 concept illustrations owned and reviewed inside SkillPilot (7).
External learning materials	Optional packages under `content/` (5)	Provider-specific textbooks, simulations, videos, and exercise collections referencing stable canonical `goalIds` from outside the core graph.

Normative rules:

- **No new external provider links on canonical goals:** Canonical `SkillLandscape` JSON files must not add third-party textbook, video, or practice-platform links in `resourceLinks`. Earlier practice-link migration work remains historical provenance evidence only and is never a model for re-attaching external providers to canonical nodes.
- **Privacy-safe plain URLs & honest coach boundary:** Any surfaced link must be a plain, tracking-free URL without learner or session identifiers. AI coaches may mention offered links as optional reading, but must **never** claim to have fetched or read external web pages they did not inspect.

5 Content Integration: Architectural Framework

This chapter establishes the binding v1.1 architectural framework (Product Owner decision of **21 September 2026**, GitHub Issue `#50`; PoC details in Physik-Libre PoC, PoC Operations, and Issue #54 Content Packages).

5.1 Three-Layer Separation and Personal Selection

```
Canonical Skill Graph (provider-neutral goalIds)
        ^
        |  referenced from outside via content/ packages
Optional Content-Link Packages (textbooks, simulations, tasks)
        ^
        |  activated per learner via ordinary SkillPilot ID
Personal Material Selection (Cockpit & Coach, default opt-in/off)
```

Core architectural invariants:

- **Provider-neutral canonical core:** Canonical landscapes never hardcode or privilege external textbook, video, or publisher URLs; content packages under `content/` point inward to stable canonical `goalIds`.
- **Ordinary SkillPilot-ID personalization:** Learners select or deselect optional content packages directly in their ordinary Cockpit profile without special activation keys, teacher accounts, or third-party logins.
- **Operational kill-switch & privacy:** Every external content package can be disabled centrally via an operational kill-switch without touching canonical landscapes or learner mastery. Outbound links remain plain, parameter-free URLs without SkillPilot IDs or tracking tokens; SkillPilot relicenses no third-party content.

5.2 Physics Pilot and Four Acceptance Criteria

The initial reference pilot binds open-licensed physics textbook sections (`https://physikbuch.schule/`) after Physics `M7` stabilization and expands to Mathematics and LEIFIphysik under Issue `#54`. Every content-package integration is judged against four binding acceptance criteria:

Criterion	Required Proof
1. Unchanged core	Disabling or removing all `content/` packages leaves canonical graphs, views, frontier logic, and mastery 100% intact.
2. Personal selection	Learners control package visibility through their ordinary SkillPilot profile without special keys or accounts.
3. Concrete utility	Active links resolve to goal-specific reading or practice sections and appear cleanly in Cockpit and Coach contexts without fabricated page summaries.
4. General viability	Adding a second provider or subject requires only a declarative `content/` package and validation pass, zero core schema changes.

6 Human-Readable Source Rationales

Every canonical learning goal in a SkillPilot skill landscape should be explainable back to official curricular evidence without requiring teachers or reviewers to inspect raw JSON ledgers or SPARQL outputs (operational commands live in Source Rationales Runbook).

6.1 Deterministic Explainability Envelope and Dual Evidence Routes

`generateGoalSourceRationales.ts` compiles a deterministic `schemaVersion: 1` JSON envelope per goal combining two complementary evidence routes:

- **Status classification (`sourceRationaleStatus`):** `classic_source_reviewed` (backed by reviewed official source extraction and mapping), `partial` (incomplete or broader topic-level coverage), or `gap` (missing source proof, surfaced honestly under `limitations`).
- **Classic source route (`classicSourceRoute`):** Binds reviewed `source-to-canonical` mappings (`exact` or `partial`), official syllabus documents, section breadcrumbs, and short statutory locator excerpts.
- **MEM SPARQL route (`memSparqlRoute`):** Cross-references external Metadata for Educational Media (`MEM`) / FWU ontology alignments (MEM SPARQL Audit) where available.

6.2 Deterministic Ranking, Delivery Modes, and Copyright Discipline

Candidate source records are ranked deterministically:

1. `exact` mappings precede `partial` mappings;
2. active/current state syllabi precede retired legacy snapshots;
3. sources matching the learner's selected jurisdiction precede nationwide or cross-state backup evidence.

In standalone curriculum package mode (11), the runtime serves generation-bound evidence via Catalog 1.2 discovery rather than hardcoded repository paths, failing closed without falling back to stale static files. Official syllabus quotes remain strictly bounded short citation locators (never full-document reproductions) to respect statutory/publisher copyright boundaries.

7 Atomic Goal Visualizations

Optional concept illustrations help learners grasp the core idea of an atomic learning goal visually without replacing independent reasoning or task solving.

7.1 Canonical Link Schema and MCP App Presentation Contract

Reviewed visualizations attach to canonical atomic goals via `resourceLinks`:

```
{
  "type": "goal-visualization",
  "resourceType": "image",
  "role": "primary",
  "skillpilotId": "<exact-canonical-goal-id>",
  "url": "/assets/goal-visualizations/<subject>/<id>/<id>.png"
}
```

In MCP-capable hosts such as OpenAI ChatGPT, `render_skillpilot_goal_visualization` serves a surface-neutral `text/html;profile=mcp-app` widget governed by a one-shot `(goalId, stateVersion)` presentation authorization and the learner toggle `showGoalVisualizations InChat`. The widget remains collapsed until the image loads cleanly, accepts bounded MCP result envelopes while using persisted widget state only as a bootstrap fallback, and retains all previously advertised content-addressed UI resource hashes unchanged.

7.2 Production Workspace, Format, and Provider Policy

Authoring prompts, candidate generation, and QA ledgers reside under `curricula/DE/Gymnasium/visualizations/` (`prompt.de.md`, `image-reconstruction-prompt.de.md`, published via `npm run deploy:assets` at 16:9 `1600x900` PNG and inspected at both `360px` mobile and `680px` desktop widths, calibrated on pilot goal `502ecaa7-cca6-5c51-a1cc-da09a7b2382c`).

Under the binding **Product Owner decision of 13 September 2026**:

- **Default `KEEP`:** Retain good existing Mathematics and Physics images regardless of which provider generated them; never regenerate retro-actively merely to unify providers.
- **Provider priority for new or defective images:** Prefer **ChatGPT/Codex image generation** in SkillPilot's friendly, abstract, clear, comic-like raster style when subject-matter accuracy and representation fit. **Nano Banana Pro** (`gemini-3-pro-image`) is an allowed alternative whenever it better serves the didactic representation — no prior failed attempt with the other provider is required.
- **Narrow SVG/programmatic exception:** Programmatically plotted diagrams or SVGs remain narrow, explicitly documented exceptions justified by exact geometric/function-plot precision, never a blanket style replacement.

7.3 CLI Workflow and Two-Stage Review Gates

Helper scripts in `app/` (`visualization:generate:nano-banana`, `visualization:plan-batch`, `visualization:check-prompt-appends`, `visualization:prepare`, `visualization:import`) separate asset generation from approval:

1. **AI QA ledger (`aiApprovedAssetSha256` / Gate V):** Records structured review labels (`accepted_pilot`, `accepted_pilot_after_regeneration`, `rejected_regenerate`, `deferred_provider_limitation`, `needs_external_review`) bound to exact file SHA-256 digests.
2. **Human publication gate (`humanApproved`, `HR-003`):** Public Standalone releases and `public`-mode GoalBooks (8, Human Review Gates) require explicit human content and rights clearance (`CC-BY-4.0` vs. provenance); generation or AI review alone never grants publication approval.

8 Learning-Goal Book and Evidence-Review Pipeline

The learning-goal book (`Lernzielbuch`) and positive understanding-evidence pipeline make every curricular-atomic goal inspectable by humans and multi-AI reviewers without bloating learner-facing goal descriptions (teacher timeline planning on top of the GoalBook is specified in Goal-Book Time Planning).

8.1 Shared BookModel, One-Goal-Per-Page Atlas, and Evidence Profiles V2

One deterministic `BookModel` (`contracts/goal-book/v1/goal-book-model-1.1.schema.json`) drives both the read-only `/lernzielbuch` WebGUI and the A4 portrait PDF:

- **Current nationwide review editions (30 Aug 2026): 792** unique curricular-atomic pages for Gymnasium Mathematics and **461** pages for Physics across all 16 German states (`publicationMode: review`, carrying QA-ledger candidate images without claiming human image approval; coach runtime and Mastery writes remain unchanged).
- **Fourteen binding invariants:** Concise canonical `title`/`description`; separate fingerprint-bound V2 evidence profiles (`contracts/goal-evidence/v2/goal-evidence-profile.schema.json` capturing `expectations`, `coverageExpectations`, `variationAxes`, and `applicationCaseBriefs`; V1 retained as archived audit history); **one physical page per ordinary curricular-atomic target goal** displaying full **Lernziel-ID**, chapter path (`canonical-composition-view-v1` + **Bundeslandspezifische Ergänzungen** with gap-free `treeOrder`), stable Kahn topological order from `requires`, clickable named PDF destinations (`goal-<uuid>`), fail-closed overflow/legibility checks, incremental Skia PDF 1.4 bookmark outlines, state-primary applicability tuples (`Bundesland -> [{stage, durationModel, courseProfile}]` — never Cartesian `G8`/`G9` cross-products), and on-demand `/lernzielbuch/<bookId>.original-sources.json` supplements.
- **Application-based recognition review (4 Nov 2026):** Assessments supporting global competence recognition (Assessment-Gated Completion) require static review against compensating-point bypasses before release.

8.2 Multi-AI Review Loop, Evidence Gates, and Moderated Feedback Custody

Quality iteration combines a fast blind multi-AI loop with strict generalization gates and a slow, privacy-minimized public feedback channel:

Scale	Levels & Meaning
Evidence (`E0`–`E5` + `NORMATIVE`)	`E0` unanchored opinion → `E1` anchored observation → `E2` reproducible minimal counterexample → `E3` independent multi-model/prompt triangulation → `E4` controlled field evidence → `E5` replicated cohort evidence.
Claim scope (`G0`–`G4`)	`G0` single case → `G1` one goal → `G2` goal archetype (requires $\geq$ `E3` across $\geq$ 2 chapters) → `G3` subject/stage (stratified sample + replication) → `G4` global Coach/Mastery rule (Math + Physics + field canary).

- **Two-round V2 campaign:** Calibrated on 16 Mathematics goals (`5`–`10`, `E`–`Q4`, pilot goal `8dd9f210-2683-5902-acab-e3be22725232`) before the 792-goal batch: (1) freeze Atlas V1, (2) blind Codex V1 review (`keep` | `revise` | `split_review` | `block`), (3) deterministic check + goal-by-goal human adjudication, (4) controlled bilingual V2 authoring + downstream lane

re-runs, (5) blind external-provider V2 review, (6) dissent-preserving synthesis. Neither AI agreement nor silence ever auto-approves canonical changes.

- **Two-phase `/lernziel-feedback` custody transfer:** Default-off in production behind `SKILLPILOT_GOAL_FEEDBACK_OPERATOR_TOKEN`; accepts privacy-minimized V2 envelopes without learner/SkillPilot IDs, contact fields, or attachments. `POST /api/operations/goal-feedback/v1/export-batches` exports deterministic batches + server-derived publication snapshots into owner-only `tmp/goal-feedback/inbox/` (`trusted-context.jsonl` separated from `untrusted-feedback.jsonl`); only after local byte/digest verification does idempotent `DELETE .../{exportId}` with `If-Match` clear live rows while retaining a content-free batch receipt.

```
npm --prefix app run quality:goal-evidence:check
npm --prefix app run build:goal-book-model -- \
  scripts/config/goal-books/de-gym-math-national-atlas.json
npm --prefix app run build:goal-book-original-sources
npm --prefix app run check:goal-book-original-sources
```

9 Canonical Gymnasium Rollout Policy

This chapter specifies the durable target architecture and rollout rules for converging German Gymnasium curricula across all 16 Bundesländer into one shared canonical skill graph per subject without duplicating competencies per state (execution plans and migration ledgers live in Canonical Gymnasium Implementation Plan and Migration Status).

9.1 Nine Core Decisions and Scope Vocabulary

1. **Canonical subject layer first:** One continuous `SkillLandscape` per subject under `curricula/DE/Gymnasium/canonical/` spanning Sekundarstufe I through Abitur readiness.
2. **Legacy continuity during transition:** Retired state trees survive as state-scoped audit/source inputs under `curricula/DE/Gymnasium/input/<STATE>/` (enforced by retirement validators in Graph Validation Rules) while compatibility mappings bridge learner progress.
3. **Minimal mapping layer:** Explicit `legacyGoalId -> canonicalGoalId` mapping files (`matchType: "exact" | "partial"`) instead of a second content universe.
4. **ISO 3166-2 state identifiers:** Metadata and filters standardize on `DE-HE`, `DE-BY`, ..., `DE-TH` (`ALL` is a query-only sentinel, never serialized).
5. **Stable coach/MCP APIs:** Backend/runtime code hides legacy-vs-canonical translation so AI coaches always see one learner state, one frontier, and one mastery model.
6. **Controlled cross-subject `requires`:** Canonical subject landscapes may author cross-subject edges directly (for example Physics goals requiring Mathematics atoms by their stable IDs) without cloning foreign subtrees; graph validators resolve these references across loaded landscapes, and `CQR-001` includes required foreign goals and their inheriting ancestors in its atomic-normal-form and transitive-minimality audit.
7. **Shared `Gymnasium (DE)` root & reviewed scope vocabulary:** One DE-level composition root exposes child subject landscapes governed by `jurisdiction` (`DE-*`), `schoolForm` (`Gymnasium`), `stage` (`SekI`, `SekII`), `durationModel` (`G8`, `G9`), and `courseProfile` (`GK`, `LK`, `GK+LK` union without duplicated IDs), backed by explicit state-scoped `.view.json` files (3).
8. **G9 Sek-I reference grid:** Canonical Sek-I authoring normalizes onto the shared `Jahrgang 5`–`Jahrgang 10` grid while preserving `G8`/`G9` source truth in provenance and view bindings.
9. **State-scoped retained inputs:** Non-canonical source extracts, Abitur blueprints (`curricula/DE/Gymnasium/input/HE/abi/`), and audit records remain strictly state-scoped under `input/<STATE>/`.

9.2 Subtree Migration State Machine and Adoption Sequence

The unit of convergence is a **didactically closed subtree** (never a forced big-bang rewrite), progressing monotonically through four states:

State	Meaning & Learner-State Contract
`1. legacy_frozen`	Existing state curriculum JSON is read-only source input; runtime/mapping fixes allowed without rewriting legacy trees.
`2. subtree_adopted`	Closed `contains` subtree reconstructed in `canonical/` with explicit `exact`/`partial` mappings and bound prerequisites.
`3. cutover_ready`	Learner-state continuity, frontier coherence, and mapping coverage verified so canonical navigation can become default.
`4. legacy_view_retained`	Canonical subtree is the primary learner-facing route; compatibility views remain temporarily until formal retirement.

Strategic adoption sequence: preserve Hessen learner continuity → establish Mathematics across Sek I and Sek II as the primary prerequisite spine → fold Bavaria and remaining states into canonical Mathematics via mappings and `.view.json` projections → widen multi-subject canonical navigation and cross-subject prerequisites.

10 MEM/FWU Roundtrip Plan

SkillPilot treats Metadata for Educational Media (MEM) and FWU-DE/lehrplan-ontologie as an adjacent semantic metadata and provenance layer rather than a competing model, acting as a practical open-source test bench for traceable AI learning-step navigation (see the operational pipeline in MEM/FWU Roundtrip Pipeline and reference paper <https://aifyer.com/ki-lernbegleitung/versions/v1.0/schulisch-verantwortete-ki-lernbegleitung-v1.0.pdf>).

10.1 Semantic Roundtrip Probe and Required Coverage

```
SkillPilot skill landscape (JSON export ZIP)
  -> MEM/FWU-first RDF/OWL bundle + hashed binary sidecars
  -> Reconstructed SkillPilot skill-landscape data (no source JSON access)
  -> Deterministic SkillPilot graph, view, and asset validation
```

Without embedding opaque JSON blobs or Base64 media literals, the semantic bundle preserves:

- Goal identity (`id`, `shortKey`, `title`, `description`), atomic/cluster classification, `weight`, `core`, and dimension metadata;
- Direct `contains` composition and didactic `requires` prerequisites;
- Program structure (`programUnits`), `goalPlacements`, competency axes (`competencyCatalog`, `competencyRefs`), and learner-facing composition views;
- Official source/provenance references plus `goal-visualization` references (`schema:ImageObject`) paired with byte-identical SHA-256 image sidecars (learner mastery state stays strictly outside public curriculum bundles).

10.2 Core-First FWU Alignment and Application Profile

Relation / Concept	Core-First RDF/OWL Representation & Constraints
Didactic `requires`	Specialized reified `LP_0000554` resource attached via `LP_0030071` (`hat Verweis`) and pointing to the prerequisite via `LP_0030072` (`verweist auf`, introduced in `FWU-DE/lehrplan-ontologie#9`). Generic `CE-Verweis` links must **never** be misread as prerequisites.
Curricular `contains`	`BFO_0000051` (`hat Teil`) only for genuine curricular part-whole relations, paired with `skillpilot:containsGoal` on every authored direct edge so transitive BFO reasoner inferences never obscure direct edges (mixed practice/exam/memory/navigation edges use only `skillpilot:containsGoal`).
Minimal application terms	`skillpilot:{LearningGoal,AtomicGoal,ClusterGoal,containsGoal,weight,core,shortKey,ProgramUnit,GoalPlacement,CompositionView,GoalVisualizationReference}` plus `schema:ImageObject` for binary sidecars.

Success requires full SkillPilot validator conformance (acyclic `requires`/`contains`, referential integrity, valid composition views), deterministic field equivalence, OWL 2 DL / structural core validation, and byte-identical image SHA-256 verification without hidden access to the original JSON input.

11 Dual Curriculum Packages: JSON Runtime and Curriculum Ontology

Every versioned curriculum release publishes **one exact subject-matter state** in two semantically equivalent representations — without dual authoring and without requiring a SkillPilot deployment to read the `curricula/` source tree (current checkpoint: Dual Package Implementation Status; operational contracts: Package Readiness, Release-Model Conformance, Dual Release Contracts, and Package Provisioning).

11.1 Seven Binding Architectural Decisions and Release Artifacts

```
Authoring sources -> Release compiler -> Normalized release model
  -> JSON runtime ZIP (<id>-<ver>.json.zip)
       -> Core-first ontology exporter -> FWU/OWL ZIP (<id>-<ver>.fwu-owl.zip)
            -> Isolated reverse compiler (bubblewrap, no source JSON)
                 -> Reconstructed JSON model -> Semantic equivalence gate
```

1. **No dual authoring:** Both ZIPs derive from one source state; fixes happen only in authoring sources.
2. **JSON ZIP is the normative runtime contract (`full-standalone-v1`):** Contains `metadata/manifest.json` (explicit artifact roles, offline `schemas/catalog.json`), `data/runtime/{catalog.json,dependency-closure.json,migration-aliases.json}`, compiled `applicability`, `canonical/`, `views/`, `cards/`, `resources/`, `mappings/`, `sources/`, `metadata/quality/`, `assets/`, and `licenses/`, operable with zero `curricula/` or `app/public` fallbacks and strict ZIP32 bounds (`3.5 GB` archive, `60,000` entries, `64 MiB` JSON/image, `128` JSON depth, `100:1`/`25:1` compression ratios).
3. **Ontology ZIP derives strictly from the finished JSON ZIP:** Contains `rdf/{declarations,runtime,landscape,views,mappings,sources,cards,assets,bundle}.nt`, `skillpilot-curriculum-profile.ttl`, `catalog-v001.xml`, pinned `ontology/{lehrplan-core.owl,shapes.ttl}`, and identical binary sidecars (validated for RDF syntax, SHACL, OWL 2 DL via ROBOT, and HermiT consistency).
4. **Semantic equivalence & shared `contentDigest`:** Governed by a closed **454-rule field-semantics registry** (`323` runtime + `130` publication normal forms + `1` legacy entry; `scalar`, `ordered-list`, `set`, `map`, `binary-reference`, `generated-non-semantic`; mapped via `fwu-core`, `skillpilot-profile`, bounded `registered-canonical-json-literal`, or `excluded-generated`). Unregistered fields fail the build; placement identities use `goalId@unitId@index`.
5. **Core-first, not Core-only:** Declarative subject profiles map required `semanticKind` values (`curricularAtomic`, `curricularArea`, `programStructure`, `practiceAssessment`, `memory`, `orientation`, `runtimeSupport`) without title guessing.
6. **Zero hidden JSON carrier:** Isolated netless `bubblewrap --clearenv` reverse compilation (`DPK-008d`) proves lossless reconstruction from RDF + sidecars alone.
7. **External release index & signing projection:** `<id>-<ver>.release.json`, `.equivalence.json`, `.provenance.json`, `.SHA256SUMS`, and `.release.json.sig` (stripping `authentication.releaseIndexSignature` before canonical hash binding) plus a signed release catalog (`staging` → `stable`, never unpinned `latest`).

11.2 Generic Runtime Decoupling, Versioning, and Phases 0–6

Four generic components decouple SkillPilot from repository paths: `CurriculumPackageRepository` (read-only lock/CAS reader), `JsonCurriculumPackageLoader` (schema/hash-verified domain builder), `CurriculumResourceResolver` (versioned `(packageId, deckId,`

`locale)` and `(packageId, resourceId)` hrefs), and `CurriculumActivationService` (quarantine extraction, dry-run validation, atomic CAS lock switch, rollback). Version axes stay orthogonal (`packageVersion`, `curriculumEdition`, `packageFormatVersion`, `runtimeContractVersion`, `supportedSkillpilotSoftware` currently `>=0.1.0 <1.0.0` for consumer `0.1.0`, `ontologyProfileVersion`, and pinned FWU Core commit/SHA), while `migration-aliases.json` distinguishes `renamed` (auto-transfers mastery) from `replacedBy`, `splitInto`, `mergedInto`, and `removed`. Licensing separates `licenseExpression`, `provenanceClass`, and `redistributionStatus` (`review-required` keeps staging candidates `not-ready-incomplete`).

Phase	Scope & Recorded Milestone Status
Phase 0	**Technically complete (`DPK-004` / `DPK-007a`):** Frozen Mathematics `0.1.0-conformance.3` (`914` ZIP entries, `757` images, 454-rule registry; intentionally `not-ready-incomplete` pending human image/rights/source-text gates).
Phase 1	**Technically complete (`DPK-005b`–`DPK-007`):** Hermetic `bubblewrap` + Corretto + Playwright consumer verified (`1,079` goals, `33,382` source mappings, `88` views, `12` decks, `757` images, zero repository-poison reads under `strace`).
Phase 2	**Technically complete (`DPK-008a`–`DPK-008d`):** Reproducible FWU-OWL bundle (`824,452` triples, SHACL/ROBOT/HermiT green) and isolated byte-identical JSON ZIP reconstruction (`911` entries, `111` logical artifacts).
Phase 3	**Next planned (`DPK-009`, `not_started`):** Unified downstream equivalence oracle, dual hermetic consumer smoke tests, isolated `JSON -> FWU` pair reproducibility, and signed external `release.json`.
Phases 4–6	Multi-subject generalization (Math → Physics embedded closure → languages → Chemistry) → multi-package production catalog/upgrades → repository split between curriculum tooling and SkillPilot software.

11.3 Runtime Contract Versions and Capabilities

A JSON runtime package selects exactly one runtime contract: its manifest names one release profile, and the catalog's `runtimeContractVersion` fixes the catalog schema and the exact capability set. Optional capabilities are not negotiated; the Java loader rejects a completion capability outside runtime 1.1 and a learning-unit capability outside runtime 1.2. Published files of an earlier contract version stay byte-identical when a new version is added beside them.

Runtime contract	Capability and release profile	Carries
1.0	none; `full-standalone-v1`	The package described in this chapter, including the calendar-style placement layer.
1.1	`assessmentCompletionV1`; `full-standalone-completion-v1`	1.0 plus completion profiles, scopes, and their reviewed rules for the assessment-completion pilot. They implement the waiver semantics that the concept revision of 4 November 2026 superseded.
1.2	`learningUnitsV1`; `full-standalone-learning-units-v1`	1.0 plus learning units: program units of kind `learningUnit`, `own` placements, coverage reviews, and `learningUnitCoverage` (Graph Definition §11.2); each review record ships unchanged with its SHA-256. No completion artifacts.

1.1 and 1.2 are siblings derived from 1.0; 1.2 is not a superset of 1.1. A build under a 1.0 or 1.1 profile of a landscape that contains learning units projects the learning-unit layer out completely

before any other processing and never touches goal objects, so its output is byte-identical to the build of the same landscape without learning units; a calendar unit below a learning unit cannot be projected and fails such a build. A 1.0 or 1.1 package that carries any learning-unit field or review record is rejected. The release compiler, the release-model validator, and the finished-package validator check the learning-unit rules of 1.2 fail-closed, and the FWU-OWL carrier asserts ownership, hierarchy, and coverage reviews as RDF relationships that the reverse compiler must reconstruct exactly. Learning units are the successor of the 1.1 completion artifacts: the 1.1 pilot keeps running (in repository mode, or from a 1.1 package for a package-mode consumer) until the runtime switches, which is a later change. The Java loader accepts 1.2 packages and the landscape model round-trips every learning-unit field, but no recognition or completion is derived from learning units yet (contract 1.1, contract 1.2).

12 Learning-Unit Design Criteria and Examples

This chapter makes the didactic purpose of learning units concrete. It explains how to draw and test a unit boundary and applies these criteria to two worked candidates: motion with constant acceleration in upper-secondary physics and number sense in grade 5 mathematics.

Status, 11 November 2026. The examples are **concept candidates**, not reviewed assessment contracts and not an approved partition of any official curriculum. This page changes no learning goals, graph edges, assessments, or access rules. The binding contracts are Learning Units (Graph Definition §11), Implicit Mastery (§13), and the separate progression and access decision. Where this chapter and those contracts differ, the contracts win.

12.1 Establish a Secure Foundation

A learning unit bundles one coherent conceptual step: learners understand a model or a subject-matter relationship and can act independently with it. The motivation explains which question becomes answerable and why the effort pays off. A unit may correspond to a familiar topic; it may equally open up a new way of thinking that connects several topics. Its scope follows this learning promise. A final application or synthesis can also form a unit; it does not need a later unit to justify its existence.

The unit's assessment, or its set of assessments, shows whether this foundation actually holds. In the regular course, material that builds on it is developed only after the required foundation has been demonstrated. Otherwise a gap is made visible, worked on in a focused way, and assessed again. An honestly identified deficit is more helpful than a pass whose apparent security later lets the learning path fail.

This does not demand flawless performance. An isolated arithmetic slip is judged differently from a missing central understanding. Ordinary grading must reliably reflect this difference: a competence that is promised but fundamentally missing must not be compensated by incidental points elsewhere. This creates no second pass verdict and no second stored outcome; the ordinary grading contract of the assessment applies unchanged.

12.2 Recognising a Meaningful Boundary

A boundary is supported by an understandable shared learning promise and by an independent application that can test the whole claimed conceptual step. A recognisable transition to later demands is another important reason; a final unit needs no successor. Subject proximity and dense prerequisite connections are indications, not an automatic cut. A sparse, transitively minimal chain can also be closely connected. Shared foundations such as equations or number sense connect many topics without turning them into one giant unit.

The following justifications are unsuitable:

- "There already is an assessment": the assessment must arise from the learning promise; its existing scope does not determine the unit.
- "The group needs at least a few goals": there is no minimum number. A single goal can be comprehensive enough; many goals can be only loosely connected.
- "The heading looks tidy": an additional parent heading creates neither coherence nor a new learning outcome.
- "Then all edges fit": prerequisites are neither invented nor removed to make a desired partition fit afterwards.

A subject-matter questionable edge is a finding of its own. Its review must not be replaced by silently narrowing the meaning of the goals involved. Even an atomically acyclic sequence can, through poor grouping, produce a contradiction between whole blocks; boundary or nesting must then be reconsidered.

The Knowledge Context of a Unit Whatever elaborates the goals of a unit addresses a learner who knows some things and does not yet know others. This knowledge is the knowledge context: a projection of the skill graph that says for every goal what it builds on, what is secured, what builds on it, and which neighbouring subject matter is not yet secured (Product Owner requirement of 13 November 2026). The knowledge context of a unit is the union of the contexts of its own goals and assessments. Nothing is written for a unit by hand.

In a unit on differentiating polynomial functions, the context secures power functions with natural exponents, their multiples and sums, and the rules the unit develops. Trigonometric, exponential, and logarithmic functions and the product and chain rules are not secured. An application in this unit, and equally the image of one of its goals, then shows a polynomial function and never $\sin x$ or e^x, however natural such an example looks to an author who knows the whole subject.

Two rules follow for the design of a unit:

1. **Every elaboration stays inside.** Tasks and sample solutions of the own assessments, memory cards, practice tasks, goal images, and the coach's examples for the unit's goals use only what the context secures and what the unit develops. Whatever leaves it is assigned to the wrong unit or is changed.
2. **The graph must suffice.** If the unit needs something that its context does not secure, the graph lacks a prerequisite, a goal reaches into later subject matter, or the boundary is wrong. That is a finding for subject review. No edge is invented and no goal text is narrowed to remove it.

The context is derived before tasks, cards, and images of the unit are written or reviewed. It is the same in every curriculum context, because the prerequisite relation is canonical; "planned for the same school year" secures nothing.

12.3 Learn, Demonstrate, and Continue Learning

The course starts with a meaningful question and a clear outlook: what can I do afterwards, and what do I need it for? Required prior knowledge is named concretely. Development and practice with feedback follow. The coach may explain, structure, and support when difficulties arise.

The subsequent assessment demands the claimed independent application. Step-by-step help that supplies exactly the decisive choice of model or reasoning can be practice, but it does not demonstrate this independence. Task variants must carry the same learning promise and must not merely repeat the practised surface.

When the evidence is missing, the concrete gap is explained, for example the confusion of two quantities. Suitable further development and practice follow, then a fresh assessment variant with equivalent content. A retry must not sidestep the problem through memorising an identical task. Valid competence evidence obtained elsewhere is not deleted by this.

The didactic principle is thereby fixed. Whether technical start locks follow from it, and how deliberate direct assessment attempts behave under such a lock, remains a separate, undecided policy; see progression and access. Direct assessment access remains the current baseline, and an attempt that is not passed records no learner result.

12.4 Main Example: Motion with Constant Acceleration (Upper-Secondary Physics)

Learning Promise and Coherence **Candidate:** "I can independently model, calculate, and explain a one-dimensional motion with constant acceleration." Learners choose an axis and signs, take initial position and initial velocity into account, and connect position, velocity, and acceleration in formulas and graphs.

The guiding question can be: "How do I describe a thrown ball from release to impact with a single model?" Free fall from rest and vertical throws are cases of the same model here. They are not split into two single-goal units merely because they carry different names. Different initial conditions make the common relationship visible.

For the time from the chosen start onwards, constant acceleration gives:

$$x(t) = x_0 + v_0 t + \tfrac{1}{2} a t^2, \qquad v(t) = v_0 + at, \qquad a(t) = a.$$

In a course where derivatives are already available, learners can use them to check and physically interpret the relationships from x to v and from v to a. This is not a blanket assumption about every entry course. The quadratic equation of motion can be developed without integral calculus or provided as a model equation; the entry described here requires no formal integral calculus.

The following simplified values illustrate the relationship for a vertical throw from release height ($x_0 = 0$, $v_0 = +20\,\mathrm{m/s}$, $a = -10\,\mathrm{m/s^2}$, no air resistance, up to the return to release height):

t in s	x in m	v in m/s	a in m/s^2
0	0	+20	−10
1	15	+10	−10
2	20	0	−10
3	15	−10	−10
4	0	−20	−10

At $t = 2\,\mathrm{s}$ the position reaches its maximum and the velocity is zero, while the acceleration remains $-10\,\mathrm{m/s^2}$. The table serves the concept review only; it is not an approved assessment task.

Entry and Continuation What is needed are the mathematical skills these tasks actually require: using terms and units and solving equations, including the quadratic equations that occur. If the chosen task demands derivatives, those must be available beforehand as well. An application visible in the physics task does not prove the complete corresponding mathematics goals across the board; their own meaning and valid evidence count for that.

The knowledge context of this example secures the description of motion in diagrams, uniform motion, and average and instantaneous velocity, together with linear functions and the reading of graphs. Vectors and motion in two dimensions, forces as causes, and derivatives and integrals are not secured. Neither are quadratic equations: no prerequisite path of the two goals provides them, although the tasks need them. That is a finding under the second rule of the knowledge context of a unit.

Uniform motion can form a sensible subject-matter entry, for example to distinguish velocity from acceleration. This creates no obligation to complete every physics unit that comes earlier in the calendar first.

Force-and-motion tasks that actually build on this, or later applications with energy and motion curves, can follow. Forces and energy are new concepts there and are not already mastered through this kinematics evidence. The candidate claims neither an official order that is binding everywhere nor a lock on all other physics topics.

What an Independent Assessment Must Reveal A possible task describes a ball thrown vertically upwards from a raised platform. Without a given solution path, learners set up the model, determine the highest point and a later impact, and explain suitable position, velocity, and acceleration graphs. Air resistance is explicitly neglected; the time interval considered ends at the first impact.

A complementary variant changes the release height or prescribes the positive axis pointing downwards. The point is whether initial conditions and signs are derived from the situation instead of being copied from a familiar sequence of numbers. Not every single task must repeat every aspect; the relevant task set must cover the complete claimed learning promise.

Observation	Meaning for the Claimed Evidence
"At the highest point $v = 0$, so $a = 0$ as well."	The central distinction is missing; the complete motion model is not yet demonstrated.
A correctly set-up model contains a single calculation error with consistent follow-up working.	Not automatically counter-evidence against understanding the model; judge it by the applicable ordinary grading rule.
After an axis change the signs change, but the predicted impact stays physically consistent.	Indication of transferable understanding rather than mere formula use.

An ordinary pass rule under which the first misconception could be fully compensated by other points does not fit the claimed evidence. Conversely, this requirement implies no duty of flawlessness. It must be implemented in the ordinary task and rubric, without a hidden additional pass condition. If the existing total-score grading cannot carry the claim, the assessment or the claimed coverage must be revised before release. The task sketches and counterexamples do not yet replace a worked-out, independently reviewed production rubric.

For the misconception at the highest point, learning continues exactly there: for instance by comparing the velocity immediately before and after, with suitable graphs and the distinction between an instantaneous value and a rate of change. A fresh situation then tests the same distinction independently.

A Larger Candidate: Kinematics Including Superposition One-dimensional motion is not yet fixed as the final unit boundary. A larger unit could carry the promise: **"I can describe the motions of this entry course, choose suitable motion models, and compose motions from components."** The step from one to two dimensions would then be an inner conceptual step up to a common thematic conclusion.

A possible course leads from representations of position, velocity, and acceleration through uniform and uniformly accelerated motion to independent components and their composition. For an idealised projectile without air resistance, horizontal and vertical motion are described separately in a fixed frame of reference and recombined at the **same time**. Independence follows from this model; it does not hold for arbitrary superposed motions.

An assessment sketch of its own could present a trajectory and two contradictory explanations of its components. Learners evaluate them, justify the time assignment, and set up their own

consistent description. For an oblique throw with non-vanishing horizontal velocity, only the vertical velocity component is zero at the highest point. Whoever transfers the statement from the vertical throw to the whole velocity vector without checking has not yet demonstrated the extended relationship. The mathematical entry competencies for components, vectors, and, where needed, trigonometric decomposition must be added accordingly.

A larger unit does not move all feedback to the end. Before the course builds on the one-dimensional model, its decisive understanding must have been assessed. The larger unit can contain a required intermediate assessment for this purpose. That assessment remains an ordinary assessment goal with a binary result; it does not automatically require a child unit or an additional stored status. A conversation with decisive solution hints does not provide the independent evidence.

Three cuts therefore need to be compared:

Candidate	Justification and Test Point
Separate unit on one-dimensional motion	Does the first new model already carry a usable conclusion of its own?
Larger kinematics unit with learning sections and several assessments	Does only the transfer to components yield the intended overall understanding, without hiding early gaps?
Kinematics with genuine child units and its own synthesis	Do the child units each have their own learning promise and conclusion that justify this additional hierarchy?

The learning sections of the second variant are not yet child units. In the third variant the existing child obligations continue to apply; a synthesis does not replace their assessments. ”All of kinematics” also needs a definite scope: whether circular motion, relative motion, or variable acceleration belong to it must be decided for the concrete course and its learning goals. The two dimensions alone do not prove completeness of the kinematics topic.

Recognition Inside a Larger Unit Under the recognition rule confirmed on 11 November 2026, an own ordinary goal of a unit is recognized only once no open own assessment or memory goal of that unit still requires it, directly or transitively. In the larger kinematics candidate, the intermediate assessment on one-dimensional motion and the final assessment on composed motion both require the one-dimensional modelling goals. Passing the intermediate assessment is an ordinary binary pass with its full feedback value. It does not yet recognize those goals implicitly while the final assessment is open; they stay open unless they have been mastered explicitly. Once the final assessment is passed as well, they are recognized.

This is a consequence, not a design target. The boundary follows the conceptual transition (Section 12.2), not a desired recognition timing; splitting or rearranging assessments or units merely to obtain earlier recognition is not permitted. If one-dimensional motion is a self-contained conceptual step with a learning promise of its own, that is a didactic argument for a child unit with its own assessment. The child’s valid contract then recognizes the child’s own goals independently of further assessments of the parent. The larger unit is complete only after all its own assessments, required cards, and required child units. No new assessment type arises; acquisition ownership, evidence, and unit completion remain distinguishable, and tasks and ordinary grading must genuinely carry the claimed application. This describes the target contract, not the implemented runtime or the quality of current assessments.

What Textbook Structures Show Textbooks are examined for how they justify coherence, prior knowledge, inner order, and conclusions. Our own drafts arise from this; tasks, texts, or chapter structures are not adopted as templates. The following first observations were recorded on 6 November 2026 and have not been re-checked for this page:

- The published assignment of **Impulse Physik Oberstufe** to Hessen (Klett, 2022, PDF pages 2–3) leads, within the first book chapter, from linear motion through free fall and two-dimensional motion to projectiles and circular motion. Forces follow in the second chapter. This shows a larger thematic bracket with inner learning steps. Publisher document
- **OpenStax Physics**, chapter 5, explains its order explicitly: vector operations prepare several two-dimensional applications. The coherence is therefore a shared method. The chapter also includes other applications; its boundary is not simply our proposed kinematics boundary. Chapter introduction
- In the section on projectile motion, component decomposition and the shared time are central. Teaching notes combine comprehension questions with a targeted return to difficulties. This is a pointer towards inner comprehension checks. Section 5.3

Our conclusion: a topic can form a larger unit and at the same time need important inner checkpoints. A book chapter provides a justifiable candidate, but not yet evidence of a complete SkillPilot application assessment. A textbook's linear reading order is not a prerequisite graph, and neither a chapter number nor a large end-of-chapter test fixes the only sensible boundary of our units and intermediate assessments. The sources cited prove neither the effectiveness of our cut nor a currently binding curriculum.

Mathematical Methods Must Be Available in Time Tangents, derivatives, or area methods may be used only if the skills they need were acquired beforehand or are explicitly developed in the unit. "Planned for the same school year" is not sufficient. For every dependent activity it must be clarified whether the method is actually available at that point. Possible solutions are timely joint subject planning, explicit development of the missing method, or a suitable different approach with an unchanged competence claim.

For the entry described here, a subject-correct route without formal integration suffices: for a linear velocity, rectangles and triangles yield the signed change of position. Areas below the time axis count negatively; at a sign change the area is split. The distance travelled must be distinguished from this. This approach claims no corresponding complete method for arbitrary velocity curves. None of this silently credits mathematics mastery or changes existing prerequisites or goal scopes.

12.5 Preliminary Comparison: Number Sense in Grade 5 Mathematics

The canonical mathematics graph contains the following five goals; they are also referenced in the Hessian G9 composition views. This inventory is not a partition of the whole year. The goal titles are given in English translation; the stable IDs identify the authored German goals.

Existing Goal (Translated Title)	Meaning to Be Preserved Completely
Explain and compare the decimal system as a place-value system	Explain the decimal place-value system, decompose numbers into place values, and compare it with a given other number system.
Represent natural numbers and locate them on the number line	Represent natural numbers in digits and words and locate them, with justification, on a scaled number line.
Order and compare natural numbers	Compare and order natural numbers with justification and use relation signs.
Round natural numbers	Round to a given or contextually appropriate place and explain the resulting approximation.
Justify the unboundedness of the natural numbers with the successor principle	Justify with the generalised successor principle that there is no largest natural number.

For comparison with the graph, the stable IDs in this order are `624764d6-becd-5f9b-ada3-0d4f9d143073`, `cf474eab-1379-4877-907e-58b0892ce734`, `9ef6c4fa-b97a-5d7a-86c1-96690f02d916`, `fe07241a-b779-5f35-a82d-7aa51ae74f42`, and `1a25ef44-f310-4c23-9ba8-44baec60d3b0`.

A possible promise reads: ”I understand the structure of the natural numbers, can switch between representations, and estimate orders of magnitude with justification.” The five goals are thus connected by one theme. The existing subject-matter paths are sparse, however: representation leads to ordering; place value leads to rounding and to unboundedness. Representation and place value additionally require the existing shared orientation entry, which is not subject-matter evidence. The five goals acquire no additional inherited subject prerequisites through this.

It is still open whether this forms a self-contained unit ”number sense” or whether connecting it with calculating with natural numbers yields the better conceptual step. Independent applications, the actual later demand, and the justifiability of the boundary must be compared. The count of five decides this question as little as the existence of two branches does.

A shopping context with counting and estimating can supply suitable applications. On its own, however, it does not demonstrate the comparison of number systems or the general successor argument. These parts must not be dropped silently in order to pass off a convenient final task as complete evidence. Under rule R1, each of the five goals would need an own assessment of the unit that requires it; a convenient final task that requires only some of them leaves the others uncovered, which is a rule violation and never a free recognition.

The existing edge from the goal ”Use multiplication and integer division as inverse operations” (`cafd6520-c4af-4109-9863-cc49ba6fad4d`) to the unboundedness goal is a reason for a subject review of its necessity, not an already decided correction. The calculation goal explicitly covers natural **and integer** numbers, signs, and the conditions of admissible division. Its scope must not be narrowed for the comparison with a unit on natural numbers.

12.6 Consistency with the Recognition Model and the Unit Validity Rules

Each atomic goal keeps one stored binary explicit result. Explicit mastery M is joined by recognition I derived from the fixed contracts of the directly owning unit: $M_{\text{eff}} = M \cup I$. This competence acts globally, also under another focus or Personal Curriculum. Neither additional stored mastery tiers nor a hidden second qualification verdict arise.

Recognition is derived per valid local contract of the owning unit. An own ordinary goal is recognized once no open own assessment or memory goal of that unit still requires it, directly or transitively, also through already mastered intermediate goals. If assessments Q1 and Q2

belong to the same unit and both require goal A, then A stays open after Q1 is passed until Q2 is passed as well. An open assessment of another, later unit never reopens or withholds an already recognized goal.

The Product Owner decision of 11 November 2026 binds this rule to five unit validity rules (Graph Definition §13.3):

- **R1:** every own ordinary content goal of a unit lies on a prerequisite path from at least one own assessment of that unit, directly or transitively; a path that leads only to a memory goal does not count. Validity condition of the unit definition, hard from the first authored unit.
- **R2:** for every own ordinary goal, at least one own assessment that requires it has reviewed evidence that its tasks and ordinary grading really test the goal. Milestone M5, with the current per-goal binding at M7.
- **R3:** an assessment claims coverage only for goals it requires directly or transitively. Same placement as R1.
- **R4:** every acquisition target has exactly one directly owning unit. Same placement as R1.
- **R5:** a later re-cut of units or later edges never grant additional recognition to existing results; the contract is versioned. Runtime contract and precondition before implicit recognition is enabled in production.

Implicit recognition therefore exists only inside units whose contract satisfies R1–R4 with reviewed coverage. Otherwise the unit's own ordinary goals stay open, and the violation keeps the curriculum from reaching the respective milestone. A goal that no assessment covers stays open and is a rule violation; it is never recognized for free. Learning units are declared in the placement layer of their skill landscape, as `programUnits` of kind `learningUnit` that own atomic goals through `goalPlacements` of relation `own` (Graph Definition §11.2). For the units authored in the canonical landscapes, `validate:learning-units` (validateLearningUnits.ts) enforces R1, R3, and the uniqueness part of R4 fail-closed in the required curriculum CI; the completeness part of R4 applies only to a landscape that declares `learningUnitCoverage` as `complete`. R2 is not yet enforced as a milestone gate: the validator only checks claimed coverage reviews (the unit's `coverageReviews`) and reports missing ones. R5 and implicit recognition itself are not yet implemented. Neither candidate in this chapter is an authored learning unit.

For unit design this means: every own ordinary goal of a candidate needs an own assessment that requires it, and a path only to a memory goal is not enough; an assessment's claimed coverage never exceeds what it requires, and a missing prerequisite path is a subject-review finding rather than a reason for a redundant edge; and each goal is owned by exactly one unit, while a parent may display child goals without owning them.

Assessments and required cards need their own explicit results. Explicit mastery of an ordinary goal does not automatically pass a separately required unit assessment. Conversely, a single atomic prerequisite does not automatically require completion of the whole foreign unit; that needs a subject-justified whole-unit prerequisite. A parent completion replaces no open child assessment; completed children pass no additional parent assessment. Validly recognized individual competence stays in force while other obligations of the unit are still open.

All effective prerequisites remain authoritative, including inheritance, clusters, and complete transitive paths. An already mastered intermediate goal must not hide the demand of a still-open local assessment. Transitive minimality stays in force.

Not adopted. An alternative recorded on the archived branch of pull request #83 as a "6 November 2026 target decision" proposed immediate recognition from the qualified coverage list of a single passed assessment, including proof by a parent or foreign unit. The Product Owner reviewed it on 11 November 2026 and did not adopt it.

12.7 Short Design Sheet

Before any implementation planning, answer for every candidate:

1. What can learners do independently afterwards, and why does it matter?
2. Which concrete entry competencies are needed, and are they available, also across subjects, before the first dependent activity?
3. What does the knowledge context of the unit secure, and what is not yet secured? Does the graph suffice for what the unit needs, and do all tasks, cards, images, and examples of the unit stay inside the context?
4. Which existing goals belong directly to the unit, which required child units does it have, and is every goal owned by exactly one unit (R4)?
5. What is the central conceptual step, and what actually builds on it?
6. Which tasks and ordinary grading rules demonstrate the whole promise? Does every own ordinary goal lie on a path from an own assessment (R1), and does at least one such assessment carry reviewed evidence for it (R2)?
7. Does every assessment claim coverage only for goals it requires (R3)?
8. Could someone pass despite lacking an essential competence?
9. Which focused further learning and fresh assessment follow such a gap?
10. Which sources and complete goal meanings justify scope and boundary?

Before planning again, the examples should make traceable why the boundaries hold up in subject terms, the assessments miss nothing essential, and a failure leads to meaningful further learning. An alternative cut must be comparable on the same questions. This sheet authorizes no implementation, no complete assignment of school years, and no technical access lock.

13 Knowledge Context: What a Learner Can Be Assumed to Know

Every piece of content that belongs to a learning goal, be it an explanation, an example, a task, a memory card, a goal image, or a sentence of the coach, addresses a learner who knows some things and does not yet know others. This chapter defines that knowledge as a projection of the skill graph and states what the graph must provide for it.

Status, 13 November 2026. Concept and Product Owner requirement. The derivation exists as a repository tool (`npm --prefix app run knowledge-context`, knowledgeContext.ts). Nothing else is implemented: no gate, no generation pipeline, and no coach uses the context. The numbers come from that tool on that day. This page changes no learning goals, graph edges, assessments, or coach behaviour.

13.1 The Requirement

The learner's knowledge context must be respected. The Product Owner requires three things:

- The knowledge context of a learning goal is derived **deterministically from data stored in the graph**. Nobody writes it by hand.
- It is **compact** enough to accompany every single step.
- It is given to **every AI step that generates or reviews content of the skill landscape**, as a constant cross-check, and to the **coach** together with the respective goal.

The occasion was content that uses subject matter the learner has not met at that point: a unit on differentiating polynomial functions must not show $\sin x$ or e^x in an application or a goal image while trigonometric and exponential functions come later. The reverse error exists as well: a task that needs quadratic equations although no prerequisite path provides them.

13.2 Definition

Let g be an atomic learning goal. Its **secured goals** $W(g)$ are the atomic goals that g requires directly or transitively under the normalized relation of the Graph Definition: inherit, expand to atoms, close transitively. The **knowledge context** of g is the following projection:

Part	Derivation	Meaning
Goal	title and description of g	what is being developed here
Builds on	the goals of $W(g)$ that no other goal of $W(g)$ requires	the entry competencies
Secured	all goals of $W(g)$, grouped by area	may be used
Leads to	the goals that require g directly	must not be anticipated
Not yet	the first goals of the same area outside $W(g)$: those that require no other such goal	the neighbouring subject matter that suggests itself

One rule of use applies to generator, reviewer, and coach alike: **what some goal of the catalog develops may be used only if that goal is secured for g.** What g itself develops is the subject of the work. What no goal of the catalog develops counts as baseline knowledge, such as everyday and primary-school knowledge, and is judged by review.

Only ordinary goals are listed. Assessment, memory, and orientation goals on a prerequisite path carry the path, but they develop no subject matter of their own. An area is a cluster directly below the subject root of the landscape, or a separate root.

The projection has two renderings with a fixed order of entries. The **core** consists of goal, builds on, leads to, and not yet; it accompanies the coach. The **full** rendering adds the secured goals; it accompanies generation and review.

The same definition covers the other kinds of nodes. An assessment or memory goal q develops nothing; its content may use $W(q)$. The knowledge context of a learning unit is the union of the contexts of its own goals and assessments; nothing is written for a unit by hand. Because the prerequisite relation is canonical, the knowledge context is the same in every curriculum context.

Nothing new is stored for the knowledge context. It is a projection of `contains`, `requires`, titles, and descriptions, as the frontier is a projection of prerequisites and mastery.

13.3 What the Derivation Shows

The tool derives the knowledge context of every ordinary atomic goal of the canonical catalog. For Mathematics and Physics:

Quantity	Mathematics, median	Mathematics, 90th percentile	Physics, median	Physics, 90th percentile
Goals	807		478	
Builds on	1	2	1	3
Secured	30	51	54	113
Leads to	1	4	1	4
Not yet	16	25	11	29
Core, characters of titles	985	1,425	700	1,529

The core is compact: roughly 250 to 450 tokens of titles. The secured goals rarely fill a whole cluster, so they cannot be shortened to cluster names; the full rendering lists them.

The second finding matters more. **The derived context is exactly as good as the edges.** For the Mathematics goal ”Ganzrationale Funktionen ableiten” the graph secures 34 goals, and none of them concerns linear, quadratic, power, or polynomial functions; ”Ganzrationale Funktionen über Term und Graph beschreiben” and ”Potenzen mit natürlichen Exponenten deuten und berechnen” appear under ”not yet”. The goal ”Ganzrationale Funktionen mit Differentialrechnung analysieren” does not have the goal on differentiating polynomial functions on its prerequisite path. The existing gates establish that `requires` is acyclic, atomic, and transitively minimal. None of them establishes that the prerequisites of a goal suffice.

In other subjects the graph models few or no prerequisites, and the context is empty there: no ordinary goal of English (89 goals), History (156), or Latin (115) has a prerequisite, and neither have 178 of the 268 goals of German or 339 of the 413 goals of Politics and Economics.

13.4 Sufficiency: The Property the Graph Must Gain

The knowledge context can bind content only where the graph is **sufficient**:

- **A goal is sufficient** if someone who has $W(g)$ and baseline knowledge can learn g as it is described.
- **An artifact is faithful** if it uses only $W(g)$ and what g develops.

A review with the knowledge context as input yields three kinds of findings:

Finding	Treatment
A needed competence is an existing goal outside $W(g)$.	Candidate for a `requires` edge. It is a pedagogical modelling decision with honest provenance and goes through subject review; transitive minimality keeps holding.
The description of the goal reaches into later subject matter.	Finding about the boundary or the placement of the goal. The text is never narrowed silently.
A needed competence is no goal of the catalog and not baseline knowledge.	Finding about a gap. No goal is invented.

The derivation is deterministic. Whether a text stays inside the context is a review by a model or a person. The context turns that review into a question with a checkable answer: which goals of the catalog does this text presuppose, and are they all secured?

Verdicts are **monotone**. An added edge only enlarges $W(g)$, so a goal that was sufficient and an artifact that was faithful remain so. Only when $W(g)$ loses a goal must the content of g be reviewed again, and a comparison of two states of the graph finds those goals deterministically.

13.5 Who Receives the Context

Consumer	Use
Generation and review of goal texts and edges	The sufficiency question for every goal; the review proposes candidate edges and never adds them itself.
Goal images	The image prompt carries the context; today it carries only the title and description of the goal. The image review records faithfulness.
Memory cards, practice tasks, and exams	The context of the assessment or memory goal bounds task, solution, and rubric.
Learning units	The context of the unit replaces any hand-written statement of its subject matter; the coverage review of an own assessment receives it.
Learning-goal book	The context of the goal bounds its page.
Coach	The core context of the active goal. The backend merges it with the learner's state: a secured goal that is not yet mastered is not assumed, and a "not yet" goal that the learner has mastered may be used.

The coach receives titles of goals and mastery states that its context already carries in other places. No chat prose flows back to the backend, and the selection stays with the backend.

13.6 Order of Introduction

1. **Derive.** A deterministic repository tool and a report per subject. No gate changes. This step exists.
2. **Review sufficiency** block by block, starting with one pilot block per subject. The share of goals with findings shows what it costs to make a subject sufficient. In the pilot of 13 November 2026 (derivative entry of Mathematics, kinematics of the introductory phase of Physics) both independent reviewers found the prerequisites insufficient for 20 of 39 goals; the entry chain of calculus secures no algebra of grades 7 to 9 (pilot record).
3. **Check faithfulness** of images, cards, and tasks, together with the design of the learning units of a block.
4. **Serve the coach** per subject once the subject is sufficient. A context from an insufficient graph would forbid what learners in fact know.

Two decisions are open: whether sufficiency becomes a milestone gate, and at which milestone; and whether titles and descriptions suffice as the stored data, or whether goals should name the terms they introduce so that texts can be screened mechanically. Both are decided with the numbers of the pilot blocks.